

Syllabus:- Ellipse & Hyperbola

SECTION - I

Multiple Choice Questions (MCQ)

This section contains **20 multiple choice** questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

- Consider the conic $ex^2 + \pi y^2 - 2e^2x - 2\pi^2y + e^3 + \pi^3 = \pi e$. Suppose P is any point on the conic and S_1, S_2 are the foci of the conic, then the maximum value of $(PS_1 + PS_2)$ is
 (1) πe (2) $\sqrt{\pi e}$ (3) $2\sqrt{\pi}$ (4) $2\sqrt{e}$
- The equation to the chord joining two points (x_1, y_1) and (x_2, y_2) on the rectangular hyperbola $xy = c^2$ is:
 (1) $\frac{x}{x_1 + x_2} + \frac{y}{y_1 + y_2} = 1$ (2) $\frac{x}{x_1 - x_2} + \frac{y}{y_1 - y_2} = 1$
 (3) $\frac{x}{y_1 + y_2} + \frac{y}{x_1 + x_2} = 1$ (4) $\frac{x}{y_1 - y_2} + \frac{y}{x_1 - x_2} = 1$
- The area of the quadrilateral with its vertices at the foci of the conics $9x^2 - 16y^2 - 18x + 32y - 23 = 0$ and $25x^2 + 9y^2 - 50x - 18y + 33 = 0$, is
 (1) $\frac{5}{6}$ (2) $\frac{8}{9}$ (3) $\frac{5}{3}$ (4) $\frac{16}{9}$
- Eccentricity of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ whose latus rectum is half of its major axis
 (1) $\frac{1}{2}$ (2) $\frac{1}{\sqrt{2}}$ (3) $\frac{1}{2\sqrt{2}}$ (4) $\frac{1}{3\sqrt{2}}$
- The locus of point of intersection of straight lines $\frac{x}{a} + \frac{y}{b} = \lambda$ and $\frac{x}{a} - \frac{y}{b} = \frac{1}{\lambda}$ where λ is variable, is a
 (1) parabola (2) ellipse
 (3) hyperbola (4) rectangular hyperbola
- An ellipse has foci $(3, 1), (1, 1)$ and it passes through point $(1, 3)$. Its eccentricity is equal to
 (1) $\sqrt{2} - 1$ (2) $\sqrt{3} - 1$ (3) $\frac{1}{2}$ (4) $\frac{1}{3}$
- A circle cuts rectangular hyperbola $xy = 1$ in the points $(x_r, y_r), r = 1, 2, 3, 4$, then
 (1) $y_1 y_2 y_3 y_4 = 1$ (2) $x_1 x_2 x_3 x_4 = -1$
 (3) $x_1 x_2 x_3 x_4 = y_1 y_2 y_3 y_4 = -1$ (4) $y_1 y_2 y_3 y_4 = 0$
- The curve represented by $x = 2(\cos t + \sin t), y = 5(\cos t - \sin t)$, (parameter) is
 (1) A circle (2) A parabola (3) An ellipse (4) A hyperbola

9. If α, β are the eccentric angles of the extremities of a focal chord of an ellipse, then the eccentricity of the ellipse is –
- (1) $\frac{\cos \alpha + \cos \beta}{\cos(\alpha + \beta)}$ (2) $\frac{\sin \alpha - \sin \beta}{\sin(\alpha - \beta)}$ (3) $\frac{\cos \alpha - \cos \beta}{\cos(\alpha - \beta)}$ (4) $\frac{\sin \alpha + \sin \beta}{\sin(\alpha + \beta)}$
10. Let the distance between a focus and the corresponding directrix of an ellipse be 4 and the eccentricity be $\frac{1}{2}$. If the length of the minor axis is p then $\sqrt{3} p$ is
- (1) 2 (2) 4 (3) 6 (4) 8
11. An ellipse of major axis 8 and minor axis 6 is drawn such that the lines $x - 3y + 2 = 0$ and $3x + y - 4 = 0$ are tangents to it, then the locus of its centre is
- (1) $x^2 + y^2 - 2x - 2y - 25 = 0$ (2) $x^2 + y^2 - 4x - 4y - 23 = 0$
 (3) $x^2 + y^2 - 2x - 2y - 23 = 0$ (4) $x^2 + y^2 - 4x - 4y - 25 = 0$
12. The equation of the circle passing through the foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$, and having centre at (0, 3) is
- (A) $x^2 + y^2 - 6y - 5 = 0$ (B) $x^2 + y^2 - 6y + 5 = 0$
 (C) $x^2 + y^2 - 6y - 7 = 0$ (D) $x^2 + y^2 - 6y + 7 = 0$
13. The length of the latus rectum of the hyperbola $16x^2 - 9y^2 - 32x + 36y - 164 = 0$
- (A) $\frac{32}{3}$ (B) $\frac{31}{3}$ (C) $\frac{29}{3}$ (D) None of the above
14. The equation of the ellipse with $e = \frac{3}{4}$, foci on y -axis, centre of the origin and passing through the point (6, 4) is
- (1) $x^2 + 2y^2 = 16$ (2) $16x^2 + 7y^2 = 688$
 (3) $16x^2 + 7y^2 = 344$ (4) none of these
15. The equation to the ellipse, whose focus is the point $(-1, 1)$, whose directrix is the straight line $x - y + 3 = 0$, and whose eccentricity is $\frac{1}{2}$ is -
- (1) $7x^2 + 2xy + 7y^2 + 10x - 10y + 7 = 0$ (2) $x^2 + 2xy + 10x - 10y + 3 = 0$
 (3) $3x^2 + xy + 10x - 10y + 3 = 0$ (4) None of these
16. If minor axis of an ellipse is equal to the distance between foci and length of latus rectum is 10, the equation of the ellipse will be-
- (1) $16x^2 + 20y^2 = 625$ (2) $x^2 + 2y^2 = 100$ (3) $16x^2 + 20y^2 = 1$ (4) $2x^2 + y^2 = 50$
17. If $P(x_1, y_1)$ is a point on the ellipse $b^2x^2 + a^2y^2 = a^2b^2$ then the area of $\Delta SPS'$ =
- (1) $ae\sqrt{a^2 - x_1^2}$ (2) $be\sqrt{b^2 - x_1^2}$ (3) $ae\sqrt{b^2 - x_1^2}$ (4) $be\sqrt{a^2 - x_1^2}$
18. If equation $(10x - 5)^2 + (10y - 4)^2 = \lambda^2 (3x + 4y - 1)^2$ represents a hyperbola, then -
- (1) $-2 < \lambda < 2$ (2) $\lambda > 2$ (3) $\lambda < -2$ or $\lambda > 2$ (4) $0 < \lambda < 2$
19. The foci of a hyperbola coincide with foci of the ellipse $\frac{x^2}{25} + \frac{y^2}{4} = 1$. If its eccentricity is 2 then its equation must be-
- (1) $\frac{x^2}{4} - \frac{y^2}{12} = 1$ (2) $\frac{x^2}{12} - \frac{y^2}{4} = 1$ (3) $\frac{x^2}{4} - \frac{y^2}{12} = -1$ (4) None of these

20. Area of the triangle formed by the lines $x - y = 0$, $x + y = 0$ and any tangent to the hyperbola $x^2 - y^2 = a^2$ is-
- (1) $|a|$ (2) $\frac{1}{2} |a|$ (3) a^2 (4) $\frac{1}{2} a^2$

SECTION-II

(NUMERICAL VALUE TYPE QUESTIONS)

This section contains Five (5) questions. The answer to each question is **NUMERICAL VALUE** with twodigit integer and decimal upto one digit.

21. For the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, distance between the foci is 10. From the point $(2, \sqrt{3})$, perpendicular tangents are drawn to the hyperbola, then find the eccentricity of the hyperbola
22. F_1 and F_2 are the two foci of the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$. Let P be a point on the ellipse such that $|PF_1| = 2 |PF_2|$ where F_1 and F_2 are the two foci of the ellipse. The area of ΔPF_1F_2 is
23. If eccentricities of hyperbolas $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and $\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$ are e and e_1 then $\frac{1}{e_1^2} + \frac{1}{e^2} =$
24. If C is the centre and A, B are two points on the conic $4x^2 + 9y^2 - 8x - 36y + 4 = 0$ such that $\angle ACB = \frac{\pi}{2}$, then $CA^{-2} + CB^{-2}$ is
25. The product of the perpendiculars drawn from the foci of the ellipse, $\frac{x^2}{9} + \frac{y^2}{25} = 1$ upon the tangent to it at the point $\left(\frac{3}{2}, \frac{5\sqrt{3}}{2}\right)$, is :

ANSWER KEY_DPP-14

1.	(3)	2.	(1)	3.	(2)	4.	(2)	5.	(3)	6.	(1)	7.	(1)
8.	(3)	9.	(4)	10.	(4)	11.	(3)	12.	(3)	13.	(1)	14.	(2)
15.	(1)	16.	(2)	17.	(4)	18.	(3)	19.	(4)	20.	(3)	21.	(1.25)
22.	(4)	23.	(1)	24.	(0)	25.	(9)						